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Are minimum nurse-to-patient staffing ratios needed in hospitals? An observational study in British Columbia, Canada

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ABSTRACT

Objective To evaluate staffing conditions, patient outcomes, quality of care, patient safety and nurse job outcomes in British Columbia (BC), Canada hospitals.

Design Cross-sectional study of 58 hospitals in BC with surveys of nurses and independent measures of patient outcomes.

Setting 58 hospitals in BC.

Participants 6685 hospital-based nurses working in a direct patient care role.

Exposures Hospital-wide and unit-specific patient-to-nurse staffing ratios derived from registered nurse reports of how many patients and how many nurses were on their unit during their last shift worked.

Main outcomes and measures Objective patient outcome measures included the Hospital Standardized Mortality Ratio (HSMR) and 30-day Readmission Rate, from 2022 to 2023 Canadian Institute for Health Information data. Nurses4All@BC provided data from 2024 using validated items on multiple measures (eg, nurse burnout, missed health breaks, intentions to leave, quality and safety measures such as culture of patient safety, quality of nursing care, missed nursing care).

Results Burnout (59.4%), missed health breaks (41.7%), job dissatisfaction (36.0%), intentions to leave (19.3%) and patient outcomes (HSMR mean 95.4, median 96.0, range 26–180; readmission rate mean 10.0%, median 9.5%, range 7.9%–13.8%) were high and varied across hospitals. 68.3% of nurses reported there were not enough staff, and 77.3% reported their workloads were unsafe for patients. 60.6% of nurses gave their hospital an unfavourable patient safety rating. More patients per nurse were associated with poorer hospital mortality and readmission rates, poorer job outcomes for nurses, more adverse events for patients, less favourable ratings of quality of care and patient safety, more missed nursing care and poorer ratings of staffing adequacy and management.

Conclusions Given the variability in staffing, quality and patient outcomes across BC hospitals, the implementation of a minimum nurse-to-patient ratio policy has the potential to improve patient care safety and retention of nurses.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ Measures of nurse staffing ratios, nurse well-being and job outcomes, and objective patient outcomes in 58 British Columbia hospitals are studied.
⇒ Measures are reported at the hospital level to enable the evaluation of variation across hospitals.
⇒ Current staffing ratios were compared relative to the minimum nurse-to-patient ratio policy.
⇒ Of the 76 hospitals that report patient outcome measures to the national database, our evaluation was limited to 58 hospitals from which obtained enough nurse survey respondents to produce reliable hospital-level estimates of staffing and other measures.
⇒ The cross-sectional design does not permit causal inferences.

INTRODUCTION

There is widespread global concern about the retention and well-being of healthcare workers. Among nurses, roughly half of those working in hospitals are experiencing high burnout, and one in five nurses plans to leave their hospital within a year.[1] In Europe, the WHO published its Time to Act resolution citing the retention of healthcare workers as the highest priority for action.[2] In the USA, the National Academy of Medicine launched the Action Collaborative on Clinician Well-Being and Resilience in 2017, documenting the consensus that burnout among healthcare workers is a threat to retention and patient safety and in 2024 established a National Plan for Health Workforce Well-Being.[3 4] The scientific evidence supporting recommendations from these major reports includes decades of research and systematic reviews demonstrating a link between chronic understaffing of nurses, high burnout and intentions to leave.[5–7] Recent evidence shows that among nurses who recently left healthcare employment, the major contributing

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factors to their departure were insufficient staffing and burnout.[8]

Given the well-documented relationship between chronic understaffing, poor nurse retention and unfavourable patient outcomes, the International Council of Nurses adopted a position statement in 2019 on moving towards evidence-based safe nurse staffing.[9] Increasingly more jurisdictions globally are establishing governmentally regulated safe nurse staffing ratios. In the USA, California in 2004 implemented hospital-wide ratios, followed by Oregon in 2024. Multiple US states have similar staffing legislation proposed, and the American Nurses Association endorsed federal staffing legislation in 2023. Ireland and Scotland have policies to ensure safe nurse staffing. In Queensland, Australia, the government funded a prospective evaluation of their staffing ratio policy demonstrating that the policy improved nurse to patient staffing among the intervention hospitals, and the better staffing was associated with reductions in patient mortality, readmissions and shorter lengths of stay. Cost savings from reduced readmissions and shorter lengths of stay more than accounted for the additional costs of increasing the nursing staff.[10]

In 2023, British Columbia (BC), Canada, introduced a policy initiative as a result of a negotiated agreement between the Ministry of Health and the Nurses' Bargaining Association targeting the retention, re-engagement and recruitment of nurses including the establishment of minimum nurse-to-patient ratios (mNPRs).[11] Staffing ratios in the hospital setting are unit level rather than on a per nurse basis. Registered nurses (RNs), registered psychiatric nurses (RPNs) and licensed practical nurses (LPNs) are included in the ratio. The hospital-based mNPRs go into effect in January 2025 for the following units: medical-surgical, intensive care, high acuity/step down, focused/special care, rehabilitation and palliative care.

In this study, we evaluate the variation in nurse staffing across hospitals in BC to obtain a baseline assessment of staffing prior to the implementation of the BC policy, as well as how confident nurses are that their work sites could meet the mNPR standards. We also analyse whether and to what extent there are associations between nurse staffing and various measures including nurse job outcomes (eg, high burnout), adverse events, patient mortality and readmission rates and other measures of care quality and patient safety. The findings can inform policy implementation in BC and elsewhere and serve as an important prepolicy intervention baseline to assess the effects of the policy on staffing and patient and nurse outcomes.

METHODS

Design

This is a cross-sectional study of 58 hospitals with nursing staff serving as informants on conditions of nursing care and independent data on patient outcomes. The study was reviewed and deemed exempt by the institutional review board at the University of Pennsylvania (Protocol #855686). Prospective survey respondents were informed of the study's purpose and its voluntary nature. Completion of the survey represented informed consent. Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines for cross-sectional studies were followed.

Data

This study links two sources of data: the 2024 Nurses4All@BC survey of nurses and the 2022–2023 Canadian Institute for Health Information (CIHI) data[12] on hospital-level patient outcomes and hospital characteristics.

The Nurses4All@BC survey involved an email-based survey sent to 100% of British Columbia Nurses Union (BCNU) members designated as 'actively working' (54 374 nurses). Approximately 85% of nurses in BC are members of BCNU. Data collection occurred between 13 March and 14 July 2024, with email reminders regularly sent to non-respondents. BCNU sent several email communications to their membership encouraging them to complete the survey. 15 991 responses were received, yielding a 30.5% response rate. This research team developed and conducted the Nurses4All@BC survey, which included validated items that have been used and tested for more than two decades in international survey research.[13 14]

Nurses provided the name of their hospital, which enabled aggregation of responses from nurses working in the same hospitals, as well as enabled the linkage of nurse responses with publicly available data from CIHI which reports hospital-level patient outcome measures. CIHI measures from 2022 to 2023 were used as this was the most recent data available.

Sample

Nurses were included in the analysis if they worked in a hospital; provided the name of their hospital; identified as a RN, RPN or LPN; and were in a role that involved providing direct care to patients. Nurses working on all shifts (eg, days, nights) were invited to participate in the survey. We did not obtain information about the nurses' shift length.

Hospitals were required to be acute care non-specialty hospitals, report data on hospital type (defined in the CIHI data as: small community, medium community, large community, teaching) and patient outcome measures and have respondents from the Nurses4All@BC survey. The final analytic sample was 6685 nurses employed in 58 BC hospitals with 115 nurses, on average, reporting from each hospital. The analytic sample of 58 acute care hospitals represents 82% of acute care hospitals in BC that report CIHI data. Our sample includes 100% of the medium and large community hospitals and teaching hospitals. The sample represents 62% of small community hospitals. Of those hospitals not included in the analysis, half have fewer than 10 acute care beds.

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Variables

Patient-to-nurse staffing. The primary explanatory variable was nurse-to-patient staffing, which was derived from nurse reports of how many patients were on their unit during their last shift worked, and a separate question asking how many nurses (RNs, RPNs, LPNs) were on the unit during the last shift worked. Responses were obtained from nurses working on day shift and night shift in each of the study hospitals, since nurse-to-patient staffing varies by shift. For each nurse, the ratio of patients to nurses was computed, and then the average ratio among nurses in the same units in the same hospitals was calculated to produce a hospital-level measure, an approach consistent with prior literature.[5 10 15] The patient-to-nurse staffing measure is a single measure per hospital and averages the responses of both day and night shift nurses in order to get a comprehensive representation of staffing during a hospitalisation. Patient-to-nurse staffing is reported as a hospital-wide measure, as well as for each of the acute care units with specified policy thresholds (ie, adult medical-surgical, adult intensive care units online supplemental eFigure 1, high acuity/stepdown units online supplemental eFigure 2, emergency departments online supplemental eFigure 3, special/focused care units online supplemental eFigure 4). Additional measures of staffing included nurse responses to the statements 'nurse workloads are unsafe for patients' and 'enough staff to get the work done'. Response options ranged from 'strongly agree' to 'strongly disagree' on a 4-point Likert scale. An operational failure measure[16] about staffing was used to define the proportion of nurses who reported their work is frequently delayed or interrupted by insufficient staff.

Nurse job outcomes. The primary nurse outcome measures included nurse job outcomes, such as high burnout, severe moral distress, job dissatisfaction and intent to leave their hospital. High burnout was defined as a score ≥27 on the Emotional Exhaustion Subscale from the Maslach Burnout Inventory, consistent with prior studies.[17] Nurses were asked to respond to the question 'How much moral distress do you experience related to your work?'. Response options included 'severe', 'moderate', 'mild', and 'none'. Nurses were also asked to respond to the question 'How satisfied are you with your primary job?'. Response options ranged from 'very satisfied' to 'very dissatisfied' on a 4-point Likert scale. Responses of 'a little dissatisfied' and 'very dissatisfied' were used to characterise nurses with job dissatisfaction. Intent to leave was derived from the question 'Do you plan to be with your current worksite 1 year from now' with response options of 'yes' and 'no'. Nurses who responded 'no' were considered nurses who had an intention to leave.

Patient outcomes. Patient outcome measures were obtained from CIHI data and reported at the hospital level. Hospital mortality was measured using the Hospital Standardized Mortality Ratio (HSMR). HSMR is an in-hospital mortality rate (risk adjusted for patient age, sex, diagnosis, length of stay, comorbidities and admission status). Details about the mortality indicator can be found elsewhere;[18] in short, hospital HSMRs are meant to be interpreted relative to a peer group or national/provincial average to indicate whether a particular hospital performs worse than or better than the comparator. Higher HSMRs indicate higher risk-adjusted mortality rates, indicating that the observed mortality rate was higher than the expected. The 30-day readmission rate measures the hospital's risk-adjusted rate of urgent readmissions within 30 days of discharge for care related to: obstetrics, paediatrics, surgical or medical.

Secondary outcomes. The Nurses4All@BC data provided information on several additional outcomes, including nurse-reported adverse events, patient safety and quality of care, missed nursing care, issues with staffing and management, and the nurse work environment. From a list of adverse events, nurses were asked to select how often they occur in their work setting: never, a few times a year, monthly, weekly and daily. The Agency for Healthcare Research and Quality Culture of Patient Safety questionnaire[19] and a patient safety grade measure were used to assess patient safety. From a list of 10 nursing care activities, nurses were asked to select which of the following were necessary but left undone due to a lack of time during their last shift worked.[20] Additional items included yes/no response options to questions about health breaks on the last shift worked: 'Did you receive all your entitled uninterrupted meal and rest breaks?', 'Did you have to increase your patient assignment so another nurse could get their meal/rest breaks?' and 'Did you miss a break because there was not adequate staffing to provide coverage for your patient assignment?'. A single-item measure was used to rate the nurse work environment,[21] 'How would you rate the overall work environment of your primary job?'. Response options range on a 4-point Likert scale from 'excellent' to 'poor'. Responses of 'poor' and 'fair' were combined in the analysis.

Data analysis

Nurse survey responses on each variable were evaluated for normality and determined to be normally distributed. Nurse survey responses were aggregated within hospitals to create hospital means, medians, standard deviations (SDs) and ranges for each of the measures. Bar charts display the variation in patient-to-nurse staffing across hospitals. In each bar chart, black dots indicate how many patients the nurses within that hospital said they could safely care for at a time. Beta coefficients and 95% CIs were obtained from linear regression models which modelled the association of one additional patient per nurse on the outcome measures. Models accounted for hospital type. Data analyses were performed using STATA V.17.0. P values less than 0.05 were considered statistically significant.

Patient and public involvement

Patients and/or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

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RESULTS

Across the 58 hospitals, the average hospital had 59.4% of nurses with high burnout (median 60.2%; SD 13.6; range 14.3%–90%; table 1). 12.3% of nurses reported severe moral distress, 31% would not recommend their hospital as a good place to work and 36% were dissatisfied with their job. The most common areas of job dissatisfaction were dissatisfaction with: feeling valued by the employer (61%), salaries/wages (51.4%) and opportunities for advancement (47.5%). Nearly one in five nurses reported intending to leave their hospital within a year (19.3%). The most cited reasons for planning to leave were inadequate nurse staffing (66.6%), burnout (61.9%) and dissatisfaction with management/administration (50.2%) (online supplemental eTable 1).

Avoidable readmissions were the most frequently reported adverse event, with 58.1% of nurses, on average, reporting that avoidable readmissions occur at least monthly (median 57.1%; SD 15.3%; range 30%–88.9%; table 1). In the average hospital, more than half of nurse respondents reported the following adverse events occurred at least monthly: complaints from patients and their families, use of physical restraints and nurses experiencing emotional abuse. The mean inpatient mortality rate (HSMR) among the 58 study hospitals was 95.4 but ranged substantially (median 96.0; SD 27.3; range 26–180; table 1). Nearly 10% (9.9%) of patients in the average hospital were readmitted within 30 days of discharge (median 9.5%; SD 1.3%; range 7.9%–13.8%).

Patient safety, care quality, staffing and management issues across the 58 study hospitals are described in table 2. On average, 60.6% of nurses gave their hospital an unfavourable grade on patient safety (median 64.4%; SD 16.3%; range 14.3%–88.9%). More than one-third (36.8%) of nurses rated the quality of nursing care as fair to poor, and 32.0% would not recommend where

Table 1 Nurse job outcomes, adverse events and patient outcomes in 58 British Columbia hospitals

		Mean and range across hospitals, %			
Nurse job outcomes	**Mean**	**Median**	**SD**	**Min**	**Max**
High burnout	59.4	60.2	13.6	14.3	90.0
Severe moral distress	12.3	11.6	7.1	0.0	30.0
Would not recommend worksite as a good place to work	31.0	33.8	14.9	0.0	67.8
Job dissatisfaction	36.0	35.7	10.7	9.1	58.3
Dissatisfied with the following aspects					
Valued by employer	61.0	58.1	13.3	15.8	94.1
Salary/wages	51.4	53.2	10.4	14.3	81.8
Opportunities for advancement	47.5	46.6	11.0	18.2	90.9
Control over schedule	46.8	46.3	11.0	15.8	70.6
Work/life balance	42.3	42.8	11.7	7.1	63.6
Independence at work	17.1	17.5	8.1	0.0	41.7
Intent to leave current worksite within 1 year*	19.3	18.8	8.1	0.0	41.7
Nurse-reported adverse events, occurring at least monthly:	**Mean**	**Median**	**SD**	**Min**	**Max**
Avoidable readmissions	58.1	57.1	15.3	30.0	88.9
Complaints from patients and their families	55.3	57.9	15.1	10.0	85.7
Use of physical restraints (vest or limb)	53.3	57.6	19.7	0.0	84.8
Nurses experience emotional abuse	52.5	53.8	11.3	33.3	88.9
Work-related physical injuries to nurses	46.4	49.3	14.2	6.7	73.0
Nurses experience physical violence	44.4	45.8	13.6	10.0	72.7
Patient falls with injury after admission	43.0	42.4	15.2	16.7	80.0
Healthcare associated infections	37.1	39.0	15.8	0.0	66.7
Pressure ulcers developed or worsened after admission	34.9	33.3	17.3	0.0	70.4
Patient received wrong medication or dose	28.5	28.3	13.7	0.0	66.7
Patient outcomes	**Mean**	**Median**	**SD**	**Min**	**Max**
Hospital Standardized Mortality Ratio	95.4	96.0	27.3	26.0	180.0
30-day readmission rate, %	9.9	9.5	1.3	7.9	13.8

Data obtained from 6685 direct care hospital nurses in 58 hospitals, with percentages reported at the hospital level.
*online supplemental eTable 1 presents the primary reasons for intending to leave current hospital within 1 year.

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Table 2 Indicators of patient care quality, safety, staffing and management issues in 58 British Columbia hospitals

		Mean and Range across Hospitals, %			
Patient safety and care quality	**Mean**	**Median**	**SD**	**Min**	**Max**
Unfavourable patient safety grade	60.6	64.4	16.3	14.3	88.9
Culture of Patient Safety Items, summary (i-vii)	43.0	44.6	9.9	15.7	60.2
(i) Nurses feel like their mistakes are held against them	47.2	47.3	13.1	10.0	72.7
(ii) Important patient care information is often lost between shifts	45.8	46.9	12.8	0.0	70.0
(iii) Things 'fall between the cracks' when transferring patients	62.7	65.6	15.4	9.1	88.9
(iv) Nurses do not feel free to question decisions/actions of those in authority	43.6	47.0	14.8	0.0	72.7
(v) Do not discuss ways to prevent errors from happening again	19.2	17.9	10.4	0.0	50.0
(vi) Not given feedback about changes put into place based on adverse event reports	36.5	36.3	15.0	10.0	78.6
(vii) Actions of management show patient safety is not a top priority	45.5	46.9	16.3	0.0	85.7
Quality of nursing care (poor/fair)	36.8	41.0	14.8	0.0	58.8
Would not recommend your hospital to friends/family needing care	32.0	33.7	14.1	7.0	58.8
Less than confident patients can manage care after discharge	70.6	70.1	10.8	50.0	90.9
Missed Nursing Care	**Mean**	**Median**	**SD**	**Min**	**Max**
Comfort/talk with patients	63.0	63.3	11.0	28.6	90.0
Ambulation or range of motion	54.7	53.3	13.7	27.3	100.0
Teach/Counsel patients and families	49.1	48.7	11.1	21.4	80.0
Oral/skin care	45.1	44.1	14.9	7.1	90.0
Prepare patients and families for discharge	30.4	29.8	10.1	0.0	76.5
Patient surveillance and assessment	23.9	22.4	10.1	6.2	48.0
Coordinate patient care	21.1	20.7	8.2	0.0	40.0
Administer medications on time	20.6	20.2	10.7	0.0	50.0
Treatments and procedures	10.2	10.5	6.0	0.0	21.4
Pain management	8.9	8.1	6.7	0.0	30.0
Staffing and Management Issues	**Mean**	**Median**	**SD**	**Min**	**Max**
Less than confident that management will act to resolve problems in patient care that nurses identify	79.3	83.3	17.1	30.0	100.0
Nurse workloads are unsafe for patients	77.3	79.4	12.7	26.7	100.0
Not enough staff to get the work done	68.3	71.7	17.6	20.0	100.0
Nurses work frequently delayed/interrupted by insufficient staff	63.6	65.0	17.9	20.0	88.9
Not at all confident hospital will be able to implement ratios	66.9	70.2	20.3	11.1	100.0
Nurse increased patient assignment so another nurse could get health breaks	68.4	66.7	11.7	35.7	100.0
Nurse did not receive all entitled health breaks on last shift	47.4	47.3	17.4	0.0	82.1
Nurse missed a health break because there was not adequate staffing to cover patient assignment on last shift	41.7	40.5	16.9	0.0	75.0
% of nurses working more than scheduled hours on last shift	12.6	10.9	7.8	0.0	36.4
Nurse work environment (poor/fair)	51.2	53.9	16.1	12.5	76.5

they work to family or friends needing care. Most nurses (70.6%) were less than confident their patients could manage their care after discharge. On average, the most frequently missed nursing care tasks were: comforting/talking with patients (63.0%), ambulation/range of motion (54.7%), teaching/counselling patients and families (49.1%) and performing oral/skin care (45.1%).

The majority of nurses identified issues with staffing and management (table 2). 68.3% of nurses reported there were not enough staff to get work done, 77.3% of nurses reported their workloads were unsafe for patients and 63.6% of nurses reported that their work was frequently interrupted or delayed because of insufficient staff. The majority of nurses (66.9%) were not at all confident

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that their hospital would be able to implement the BC mNPR policy. In addition to understaffing issues, 47.4% of nurses said they did not receive all their entitled breaks during their last shift, 68.4% increased their patient assignment so another nurse could get their break, and 41.7% of nurses missed their break because there was not adequate staffing to cover their patient assignment. 12.6% of nurses reported working more than their scheduled hours during their last shift. 79.3% of nurses said they were less than confident that their management can act to resolve problems in patient care that nurses identify; and over half of nurses (51.2%) gave their hospital nurse work environment a fair/poor rating (table 2).

Figure 1 depicts the variation in average patient-to-nurse ratios across 58 BC hospitals. Panel A of figure 1 depicts hospital-wide staffing ratios; panel B depicts staffing ratios on medical-surgical units. Additional figures are provided in the Supplement for other hospital units: adult CUs (ICU), high acuity/stepdown units, focused/special care units and the emergency department. The height of each bar represents the actual average number of patients per nurse within the hospital, while the black dot represents the average number of patients the nurse respondents reported they could safely care for. The red line (figure 1, panel B) indicates the mNPR policy for medical-surgical units. There is considerable variation in actual staffing across hospitals, but far less variation in the numbers of patients nurses report they could safely care for. For example, the average medical-surgical nurse cares for five patients at a time (range: 1.8–7.1) and reports they could safely care for 4.1 patients (range: 3.0–5.2). We also observed variation in medical-surgical staffing between day shift and night shift, with night shift nurses caring for one more additional patient at a time, on average (online supplemental eTable 2).

Regression coefficients and p values reported in tables 3 and 4 demonstrate that for nearly all outcomes evaluated, one additional patient per nurse is statistically significantly associated with poorer job outcomes for nurses, adverse events, poorer patient outcomes, less favourable ratings of quality of care and patient safety, more missed nursing care and poorer ratings of staffing adequacy and

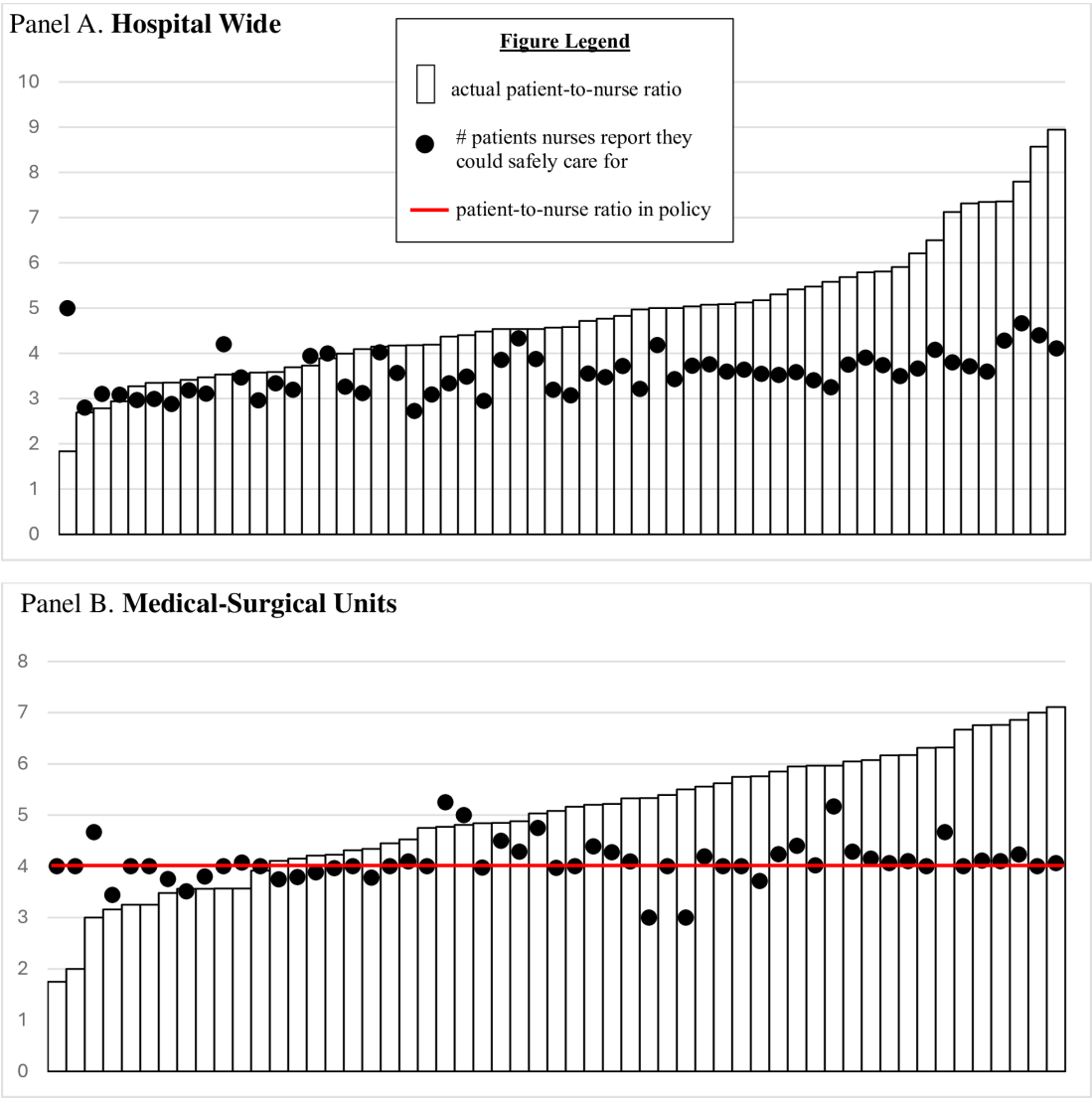

Figure 1 Variation in patient-to-nurse staffing ratios across 58 British Columbia hospitals.

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Table 3 Ordinary least squares linear regression models of the effect of one additional patient per nurse on outcomes and adverse events

	Patient to Nurse Staffing	
Nurse job outcomes	**B (95% CI)**	**P value**
High burnout	4.69 (2.09, 7.29)	**0.001**
Severe moral distress	1.91 (0.51, 3.31)	**0.008**
Would not recommend hospital as a good place to work	3.93 (1.26, 6.61)	**0.005**
Job dissatisfaction	2.14 (-0.07, 4.35)	0.058
Intent to leave current hospital within 1 year	2.18 (0.58, 3.77)	**0.008**
Adverse events, occurring at least monthly	**B (95% CI)**	**p Value**
Avoidable readmissions	3.92 (1.15, 6.68)	**0.006**
Complaints from patients and their families	4.96 (2.16, 7.77)	**0.001**
Use of physical restraints (vest or limb)	2.10 (-1.07, 5.27)	0.190
Nurses experience emotional abuse	3.30 (1.00, 5.60)	**0.006**
Work-related physical injuries to nurses	3.76 (1.36, 6.16)	**0.003**
Nurses experience physical violence	3.62 (0.95, 6.29)	**0.009**
Patient falls with injury after admission	6.25 (3.60, 8.89)	**<0.001**
Healthcare associated infections	5.65 (3.11, 8.18)	**<0.001**
Pressure ulcers developed or worsened after admission	5.55 (2.19, 8.90)	**0.002**
Patient received wrong medication or dose	4.99 (2.42, 7.56)	**<0.001**
Patient outcomes	**B (95% CI)**	**p Value**
Mortality ratio (HSMR)	7.03 (1.45, 12.61)	**0.0146**
30-Day readmission rate	0.30 (0.05, 0.55)	**0.0209**

Bold p values indicate statistical significance. All models include adjustments for hospital type (small/medium/large community, teaching)
HSMR, Hospital Standardized Mortality Ratio.

management. For example, one additional patient per nurse in the average nurse's workload is associated with a nearly five percentage-point increase in the percent of nurses with high burnout (B 4.69; 95% CI 2.09 to 7.29; p 0.001; table 3), a 4 percentage-point increase in nurses who would not recommend their hospital as a good place to work (B 3.93; 95% CI 1.26 to 6.61; p 0.005) and a 2 percentage-point increase in nurses intending to leave their hospital (B 2.18; 95% CI 0.58 to 3.77; p 0.008).

More patients per nurse was also associated with poorer outcomes for patients in terms of higher percentages of nurses reporting adverse events, higher mortality

Table 4 Ordinary least squares linear regression models of the effect of one additional patient per nurse on patient care quality, safety, staffing and management issues

	Patient to nurse Staffing	
Patient safety and care quality	**B (95% CI)**	**P Value**
Unfavourable patient safety grade	4.07 (0.92, 7.22)	**0.012**
Culture of patient safety items, summary (i–vi)	3.38 (1.61, 5.14)	**<0.001**
(i) Nurses feel like their mistakes are held against them	−0.19 (−2.73, 2.35)	0.880
(ii) Important patient care information if often lost between shifts	5.54 (3.28, 7.80)	**<0.001**
(iii) Things 'fall between the cracks' when transferring patients	4.56 (1.70, 7.42)	**0.002**
(iv) Nurses do not feel free to question decisions/actions of those in authority	3.29 (0.54, 6.05)	**0.020**
(v) Do not discuss ways to prevent errors from happening again	3.86 (1.97, 5.74)	**<0.001**
(vi) Not given feedback about changes put into place based on adverse event reports	3.62 (0.73, 6.52)	**0.015**
(vii) Actions of management show patient safety is not a top priority	2.90 (−0.38, 6.18)	0.082
Quality of nursing care (poor/fair)	4.69 (2.26, 7.13)	**<0.001**
Would not recommend your hospital to friends/family needing care	3.50 (1.16, 5.84)	**0.004**
Less than confident patients can manage care after discharge	2.47 (0.24, 4.71)	**0.031**
Missed nursing care	**B (95% CI)**	**p Value**
Comfort/talk with patients	4.81 (2.83, 6.79)	**<0.001**
Ambulation or range of motion	4.23 (1.55, 6.90)	**0.003**
Teach/counsel patients and families	3.82 (1.64, 5.99)	**0.001**
Oral/skin care	4.29 (1.40, 7.17)	**0.004**
Prepare patients and families for discharge	2.88 (0.87, 4.89)	**0.006**
Patient surveillance and assessment	1.90 (−0.23, 4.02)	0.079
Coordinate patient care	2.55 (0.91, 4.19)	**0.003**
Administer medications on time	2.21 (0.00, 4.43)	0.050
Treatments and procedures	0.92 (−0.28, 2.12)	0.130
Pain management	1.78 (0.43, 3.13)	**0.011**
Staffing and management issues	**B (95% CI)**	**p Value**
Less than confident that management will act to resolve problems in patient care that nurses identify	3.15 (−0.26, 6.57)	0.069

Continued

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Table 4 Continued

	Patient to nurse Staffing	
Patient safety and care quality	B (95% CI)	P Value
Nurse workloads are unsafe for patients	4.17 (1.69, 6.66)	**0.001**
Not enough staff to get the work done	4.51 (1.16, 7.87)	**0.009**
Nurses work frequently delayed/ interrupted by insufficient staff	5.75 (2.45, 9.05)	**0.001**
Not at all confident hospital will be able to implement ratios	6.61 (2.66, 10.56)	**0.001**
Nurse increased patient assignment so another nurse could get breaks	3.20 (0.91, 5.49)	**0.007**
Nurse did not receive all entitled breaks on last shift	5.37 (1.93, 8.81)	**0.003**
Nurse missed health break because there was not adequate staffing to cover patient assignment on last shift	5.10 (1.17, 8.49)	**0.004**
% of nurses working more than scheduled hours on last shift	1.88 (0.25, 3.50)	**0.025**
Nurse work environment (poor/ fair)	3.41 (0.50, 6.31)	**0.023**

Bold p values indicate statistical significance. All models include adjustments for hospital type (small/medium/large community, teaching)

and higher readmissions. For example, each additional patient per nurse is associated with a 7 percentage-point increase in the HSMR (B 7.03; 95% CI 1.45 to 12.61; p 0.0146) and a 0.3 percentage-point increase in a hospital's 30-day readmission rate (B 0.30; 95% CI 0.05 to 0.55; p 0.0209; table 3). Finally, except for a few quality indicators, we found that more patients per nurse were associated with greater percentages of nurses reporting worse quality of care and safety for patients, including more missed nurse care, more reports of staffing inadequacies, missing entitled breaks and poorer nurse work environments (table 4).

DISCUSSION

Public policies establishing minimum safe nurse staffing requirements in hospitals remain hotly contested by powerful stakeholders in healthcare. Resulting debates are difficult to adjudicate due to the absence of objective local data from the impacted jurisdiction establishing that a problem exists that is in the public's interest to solve through a policy intervention. The intent of this study is to provide data answering the question of whether there is a problem that adversely affects the public and whether there is evidence that the proposed solution might contribute to solving the problem. Our study documents significant variation in nurse staffing across hospitals in BC and shows that this variation is associated with preventable adverse consequences for retaining nurses and providing safe and effective hospital care to patients. Moreover, with detailed data on almost all hospitals in the province, we show that hospitals with better nurse staffing have better outcomes, providing confidence that a public policy setting minimum nurse staffing requirements is likely to result in public benefits. By establishing a baseline measure of hospital staffing and baseline nurse and patient outcomes, it will be possible in the years to come to determine whether the new staffing policy is associated with improved outcomes.

In a representative sample of 58 BC hospitals, we found high levels of nurse burnout, job dissatisfaction, intentions to leave, moral distress, poor ratings of patient safety and quality of care, and objective measures of hospital mortality and readmissions to be associated with the current variation in nurse-to-patient staffing ratios. We found that the average medical-surgical nurse cares for five patients at a time, but nurses in some hospitals care for, on average, more than seven patients at a time. Nurses on medical-surgical units report that the average number of patients they could safely care for is 4.1, which is consistent with BC's mNPR policy for medical-surgical units. We documented very similar findings in BC regarding the impact of nurse variation in staffing on patient mortality as previous research has documented in the US[15] and in Europe[14]—every one patient increase in the workload of inpatient hospital nurses is associated with a 7% increase in mortality.

High proportions of nurses in BC, like those in many other jurisdictions, are burned out and intending to leave their hospitals, thus contributing to costly nurse turnover and the possibility of a nurse shortage. We found that the primary reasons nurses intend to leave their hospitals within the year are modifiable and include inadequate nurse staffing (66.6%), burnout (61.9%), dissatisfaction with management/administration (50.2%) and not feeling valued by their worksite (48.8%). These findings signal an important opportunity to bend the curve on high nurse turnover, if hospitals can adhere to the mNPRs and successfully recruit and retain enough nurses at the bedside.

Despite the anticipated benefits that could be gained from adhering to mNPRs, the majority of nurses (67%) working in BC hospitals express low confidence that their hospitals will implement the mNPRs. Follow-up evaluations of staffing conditions across BC hospitals will be essential to understanding whether hospitals have been successful in implementing the mNPRs.

Our study is not only important to BC but to other provinces in Canada currently considering similar legislation. Canada's most populated province, Ontario, is currently debating the merits of a similar policy, and the results of this baseline study providing clear evidence of a potentially solvable problem may inform that debate. After a similar baseline study[22] in Queensland, Australia, followed by implementation of mandated minimum hospital nurse staffing and a follow-up study documenting

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improved hospital nurse and patient outcomes,[10] most other Australian jurisdictions have taken actions to adopt similar minimum hospital nurse staffing policies demonstrating the value of evidence in informing policy solutions to long-standing problems.

Limitations

This study evaluated 58 of the 76 BC hospitals that report patient outcome measures to CIHI. Hospitals not included in the sample tended to be small hospitals, some with as few as five acute care beds, which made it difficult to obtain enough direct care nurse survey respondents to produce reliable hospital-level estimates of staffing and other measures. The study was cross-sectional and therefore cannot infer causal relationships; however, research shows that results from observational and panel studies on the outcomes of hospital nurse staffing variation are very similar.[23] The response rate of 30.5% is high by current experience with email surveys.[24 25] Research on similar topics using a double sample of respondents including over 90% of original non-respondents found no response bias that impacts the variables of interest in this paper.[13] Sampling only BCNU members and funding of the study by BCNU could be considered a limitation; however, 85% of BC nurses belong to BCNU, so its membership is a reasonable sampling frame and BCNU had no input into the design of the study, interpretation of findings, or the decision to publish the findings. To the extent there was missingness on data items on the Nurse4All@BC survey, it occurred at the nurse level. Our analysis was conducted at the hospital level—and thus there was no missingness at the hospital level. Across all the survey items studied, the missingness at the nurse level ranged from 0%–17%. We did not make any attempt to impute missing data.

CONCLUSIONS

Nurses in BC hospitals are caring for more patients at a time than they believe is safe. Nurses identifying safe staffing levels in their hospitals are, for the most part, consistent with the staffing ratios set in the mNPR policy. Consistent with decades of evidence generated in other jurisdictions globally, we find that fewer patients in a nurse's workload was associated with better patient outcomes, fewer adverse events, better quality and safety of care, and better job outcomes for nurses including lower burnout and intentions to leave. If hospitals across adhere to the new nurse staffing policy requirements, they are likely to see improvements in outcomes for patients and nurses.

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